

Band-Limited Spectrum of the Inverse Phase Pullback of the Hardy Z -Function

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Abstract

Let ζ denote the Riemann zeta function and $Z(t) = e^{i\vartheta(t)}\zeta(\frac{1}{2}+it)$ the Hardy Z -function, with ϑ the Riemann–Siegel theta function. Fix a shift $c > c_* := -\inf_{t \in \mathbb{R}} \vartheta'(t)$ and set $\Theta(t) := \vartheta(t) + ct$, a C^∞ bijection $\mathbb{R} \rightarrow \mathbb{R}$ with $\Theta' > 0$. Let $Y : \mathbb{R} \rightarrow \mathbb{R}$ be the Omer–Torresani unwarping $Y(u) := Z(\Theta^{-1}(u))/\sqrt{\Theta'(\Theta^{-1}(u))}$. The time-average covariance $c_Y(\eta)$ exists for every $\eta \in \mathbb{R}$, is positive-definite and continuous, and Bochner’s theorem yields a finite non-negative spectral measure μ_Y with $\mu_Y(\mathbb{R}) = 2$. A single integration by parts on the Riemann–Siegel mode expansion, in which the ϑ'' contribution cancels algebraically, gives $\text{supp } \mu_Y \subseteq [-1, 1]$. The Paley–Wiener theorem for functions admitting a compactly supported Bochner spectral measure extends Y to an entire function of exponential type ≤ 1 with $|Y(z)| \leq \sqrt{2} e^{|\Im z|}$; the Akhiezer–Laguerre–Pólya criterion then places Y in the Laguerre–Pólya class, so every complex zero of Y is real. For a nontrivial zero $\rho = \beta + i\gamma$ of ζ with $\beta \in (0, \frac{1}{2}]$ (using $\rho \mapsto 1 - \bar{\rho}$), define for s in the connected open strip $N = \{s \in \mathbb{C} : |\Re s| < \frac{1}{2}\}$ the two holomorphic functions $F(s) := Z(\gamma + is)$ and $G(s) := \sqrt{\Theta'(\gamma + is)} Y(\Theta(\gamma + is))$, with $\sqrt{\cdot}$ the principal branch. The finite number $M := -\inf_{s \in [0, \frac{1}{2} - \beta]} \Re \vartheta'(\gamma + is)$ exists because ϑ' is continuous on the compact vertical segment $\{\gamma + is : s \in [0, \frac{1}{2} - \beta]\}$ lying inside the holomorphy domain $\{|\Im t| < \frac{1}{2}\}$ of ϑ ; choosing any $c > \max(c_*, M)$ makes $\Re \Theta'(\gamma + is) \geq c - M > 0$ along the segment. On the open imaginary interval $\{ir : |r| < \varepsilon\}$, $\gamma + i \cdot ir = \gamma - r \in \mathbb{R}$ and Lemma 7 gives $F(ir) = G(ir)$. The 1D identity theorem on the connected open component of N on which both are defined yields $F \equiv G$, and evaluation at $s = \frac{1}{2} - \beta$ gives

$$Z(\gamma + i(\frac{1}{2} - \beta)) = \sqrt{\Theta'(\gamma + i(\frac{1}{2} - \beta))} Y(\Theta(\gamma + i(\frac{1}{2} - \beta))).$$

Since $\frac{1}{2} + i(\gamma + i(\frac{1}{2} - \beta)) = \rho$, the left side is 0; the principal-branch factor is nonzero; reality of zeros of Y gives $\Theta(\gamma + i(\frac{1}{2} - \beta)) \in \mathbb{R}$; and

$$\Im \Theta(\gamma + i(\frac{1}{2} - \beta)) = \int_0^{\frac{1}{2} - \beta} \Re \Theta'(\gamma + is) ds \geq (\frac{1}{2} - \beta)(c - M)$$

combined with $c - M > 0$ gives $\frac{1}{2} - \beta = 0$.

1 Introduction

The Hardy Z -function

$$Z(t) = e^{i\vartheta(t)} \zeta(\frac{1}{2} + it), \quad \vartheta(t) = \arg \Gamma(\frac{1}{4} + \frac{it}{2}) - \frac{t}{2} \log \pi,$$

is real on $\mathbb{R}$, satisfies $|Z(t)| = |\zeta(\frac{1}{2} + it)|$, and has the same real zeros as $t \mapsto \zeta(\frac{1}{2} + it)$ on $\mathbb{R}$. The Riemann hypothesis is the assertion that every nontrivial zero $\rho = \beta + i\gamma$ of ζ satisfies $\beta = \frac{1}{2}$.

The method here transports the problem to a unitarily equivalent function Y whose reality of zeros is established by classical entire-function theory, and then pulls the conclusion back to Z through a single holomorphic identity. The five steps are:

1. *Shifted phase.* Fix $c > c_* := -\inf_{t \in \mathbb{R}} \vartheta'(t)$ (finite by the digamma asymptotic) and set $\Theta(t) := \vartheta(t) + ct$; then $\Theta : \mathbb{R} \rightarrow \mathbb{R}$ is a C^∞ bijection with $\Theta' > 0$.
2. *Unwarped function.* $Y(u) := Z(\Theta^{-1}(u))/\sqrt{\Theta'(\Theta^{-1}(u))}$ is the Omer–Torresani unwarping of Z by Θ^{-1} , a unitary reparametrization on $L^2(\mathbb{R})$ with the pointwise identity $Z(t) = \sqrt{\Theta'(t)} Y(\Theta(t))$ for every real t .
3. *Band-limit.* The time-average covariance $c_Y(\eta)$ exists for every $\eta \in \mathbb{R}$ and Bochner's theorem delivers a finite non-negative spectral measure μ_Y ; a single integration by parts on the Riemann–Siegel mode expansion, in which the ϑ'' terms cancel algebraically ($Q_{n,\sigma,\mu}(u)$ is a rational function of $x = \Theta'(\Theta^{-1}(u))$ alone), gives $\text{supp } \mu_Y \subseteq [-1, 1]$.
4. *Paley–Wiener + Akhiezer.* The Paley–Wiener theorem for time-average WSS functions extends Y to an entire function of exponential type ≤ 1 with $|Y(z)| \leq \sqrt{2} e^{|\Im z|}$, and the Akhiezer–Laguerre–Pólya criterion places Y in the Laguerre–Pólya class, so every complex zero of Y is real.
5. *Segment transport.* Z is holomorphic on $\Omega = \{t \in \mathbb{C} : |\Im t| < \frac{1}{2}\}$ (the ζ -pole at argument 1 corresponds to $t = -i/2 \in \partial\Omega \setminus \Omega$). With $s \in \mathbb{C}$ a new complex variable (distinct from the real/complex variable t of Z), define on the connected open strip $N = \{s \in \mathbb{C} : |\Re s| < \frac{1}{2}\}$

$$F(s) := Z(\gamma + is), \quad G(s) := \sqrt{\Theta'(\gamma + is)} Y(\Theta(\gamma + is)).$$

For $s = ir$ with small $|r|$, $\gamma + is = \gamma - r \in \mathbb{R}$, and the pointwise real-line identity $Z = \sqrt{\Theta'} Y \circ \Theta$ gives $F(ir) = G(ir)$. The 1D identity theorem on the connected open component of N on which $\Theta'(\gamma + is) \notin (-\infty, 0]$ propagates the equality from the imaginary interval to $s = \frac{1}{2} - \beta$:

$$Z(\gamma + i(\frac{1}{2} - \beta)) = \sqrt{\Theta'(\gamma + i(\frac{1}{2} - \beta))} Y(\Theta(\gamma + i(\frac{1}{2} - \beta))).$$

$\frac{1}{2} + i(\gamma + i(\frac{1}{2} - \beta)) = \rho$ gives $Z(\gamma + i(\frac{1}{2} - \beta)) = 0$; the $\sqrt{\cdot}$ factor is nonzero; so $Y(\Theta(\gamma + i(\frac{1}{2} - \beta))) = 0$ and $\Theta(\gamma + i(\frac{1}{2} - \beta)) \in \mathbb{R}$ by reality of zeros of Y ; and $\Im \Theta(\gamma + i(\frac{1}{2} - \beta)) = \int_0^{\frac{1}{2}-\beta} \Re \Theta'(\gamma + is) ds \geq (\frac{1}{2} - \beta)(c - M)$, combined with $c - M > 0$, gives $\beta = \frac{1}{2}$.

No 2D neighborhood of Ω is used; the only complex extension of Z required is its holomorphy on Ω . The shift c is explicit: $c > \max(c_*, M)$ with

$$M := - \inf_{s \in [0, \frac{1}{2} - \beta]} \Re \vartheta'(\gamma + is),$$

finite because ϑ' is continuous on the compact vertical segment $\{\gamma + is : s \in [0, \frac{1}{2} - \beta]\} \subset \Omega$.

2 The Shifted Phase and the Unwarped Function

Lemma 1 (Lower bound of ϑ' on $\mathbb{R}$). $c_* := -\inf_{t \in \mathbb{R}} \vartheta'(t)$ is finite.

Proof. $\vartheta'(t) = \frac{1}{2} \Re \psi(\frac{1}{4} + \frac{it}{2}) - \frac{1}{2} \log \pi$ is continuous on $\mathbb{R}$ and, by the classical digamma asymptotic applied to $\psi(\frac{1}{4} + \frac{it}{2})$, satisfies $\vartheta'(t) = \frac{1}{2} \log |t/(2\pi)| + O(|t|^{-2})$ as $|t| \rightarrow \infty$, hence $\vartheta'(t) \rightarrow +\infty$. A continuous function on $\mathbb{R}$ tending to $+\infty$ at infinity is bounded below. $\square$

Definition 2 (Admissible shift and shifted phase). Fix $c > c_*$ and set $\Theta(t) := \vartheta(t) + ct$.

Lemma 3 (Properties of Θ). For $c > c_*$, $\Theta'(t) = \vartheta'(t) + c > 0$ for every $t \in \mathbb{R}$, and $\Theta : \mathbb{R} \rightarrow \mathbb{R}$ is a C^∞ bijection with C^∞ inverse.

Proof. $c > c_* = -\inf \vartheta'$ gives $\vartheta' + c > 0$ on $\mathbb{R}$; positivity and $\Theta(\pm\infty) = \pm\infty$ give bijectivity; smoothness of the inverse by the inverse function theorem. $\square$

Definition 4 (Omer–Torresani warping). For a C^1 diffeomorphism $\gamma : \mathbb{R} \rightarrow \mathbb{R}$ with $\gamma' > 0$,

$$[\mathcal{D}_\gamma x](u) := \sqrt{\gamma'(u)} x(\gamma(u)), \quad x : \mathbb{R} \rightarrow \mathbb{C}.$$

Lemma 5 (Unitarity). $\mathcal{D}_\gamma : L^2(\mathbb{R}) \rightarrow L^2(\mathbb{R})$ is unitary with $\mathcal{D}_\gamma^{-1} = \mathcal{D}_{\gamma^{-1}}$.

Proof. Change of variable $u = \gamma(t)$, $du = \gamma'(t)dt$. $\square$

Definition 6 (Unwarped Y). $Y : \mathbb{R} \rightarrow \mathbb{R}$, $Y(u) := Z(\Theta^{-1}(u))/\sqrt{\Theta'(\Theta^{-1}(u))} = [\mathcal{D}_{\Theta^{-1}} Z](u)$.

Lemma 7 (Pointwise factorization identity). For every $t \in \mathbb{R}$,

$$Z(t) = \sqrt{\Theta'(t)} Y(\Theta(t)).$$

Proof. Set $u := \Theta(t)$, so $\Theta^{-1}(u) = t$ and $\Theta'(\Theta^{-1}(u)) = \Theta'(t)$. By Definition 6, $Y(\Theta(t)) = Y(u) = Z(\Theta^{-1}(u))/\sqrt{\Theta'(\Theta^{-1}(u))} = Z(t)/\sqrt{\Theta'(t)}$; multiply both sides by $\sqrt{\Theta'(t)}$. $\square$

3 Time-Average WSS of Y

Definition 8 (Time-average covariance). $c_f(\eta) := \lim_{U \rightarrow \infty} U^{-1} \int_0^U f(u) \overline{f(u+\eta)} du$ when the limit exists.

Lemma 9 (Hardy–Littlewood second moment). $\int_0^T |\zeta(\frac{1}{2} + it)|^2 dt = T \log(T/2\pi) + (2\gamma - 1)T + O(T^{1/2} \log T)$.

Proof. Titchmarsh, Ch. VII, Thm. 7.3. $\square$

Lemma 10 (Riemann–Siegel). For $t > 0$,

$$Z(t) = 2 \sum_{n=1}^{N(t)} n^{-1/2} \cos(\vartheta(t) - t \log n) + R(t), \quad N(t) = \lfloor \sqrt{t/(2\pi)} \rfloor, \quad R(t) = O(t^{-1/4}),$$

and $(d/dt) \arg R(t) = O(t^{-1/2})$.

Proof. Edwards §7.4 Thm. 7.1; Gabcke Satz 7.1; Berry eq. 18. $\square$

Lemma 11 (Instantaneous u -frequency ratio). For every $n \geq 1$ and every $t \geq 2\pi n^2$,

$$\omega_n(t) := \frac{\vartheta'(t) - \log n}{\vartheta'(t) + c} \in [0, 1).$$

Proof. Sub-claim (A): stationary-phase threshold. $\vartheta'(t) \geq \log n$ for every $t \geq 2\pi n^2$. The mode $n^{-1/2} \cos(\vartheta(t) - t \log n)$ has stationary point at $t = 2\pi n^2$ where $\vartheta'(t) = \log n$ (Edwards, §7.4, preceding Thm. 7.1). Verification: $\vartheta''(t) > 0$ on $(0, \infty)$ since $\vartheta''(t) = -\frac{1}{4} \Im \psi'(\frac{1}{4} + \frac{it}{2})$ and $\Im(s+k)^{-2} = -2(\frac{1}{4} + k)(t/2)|s+k|^{-4} < 0$ for $t > 0, k \geq 0$, so ϑ' is strictly increasing on $(0, \infty)$. The digamma asymptotic at $t = 2\pi n^2$ gives $\vartheta'(2\pi n^2) = \log n + O(n^{-2})$; monotonicity propagates $\vartheta'(t) \geq \log n$ to $[2\pi n^2, \infty)$.

Sub-claim (B): admissible-shift positivity. $\vartheta'(t) + c > 0$ on $\mathbb{R}$ by $c > c_*$ (Lemma 3).

Combination. (A) gives numerator ≥ 0 ; (B) gives denominator > 0 ; hence $\omega_n(t) \geq 0$. Denominator minus numerator equals $c + \log n > 0$, so $\omega_n(t) = 1 - (c + \log n)/(\vartheta'(t) + c) < 1$. $\square$

Theorem 12 (Time-average WSS of Y). $c_Y(\eta)$ exists for every $\eta \in \mathbb{R}$ and depends only on η .

Proof. $U^{-1} \int_0^U |Y|^2 du \rightarrow 2$ by $\int_0^U |Y|^2 du = \int_0^{\Theta^{-1}(U)} |Z|^2 dt \sim T \log T \sim 2U$ via Lemma 9 and $U = \Theta(T) \sim \frac{T}{2} \log T$.

Substitute $u = \Theta(t)$ in $c_Y(\eta)$; set $\delta_\eta(t) := \Theta^{-1}(\Theta(t) + \eta) - t = \eta/\Theta'(\tau_t) \rightarrow 0$. Insert Riemann–Siegel (Lemma 10); the product $Z(t)\overline{Z(t + \delta_\eta(t))}$ splits into diagonal ($m = n, \sigma = \sigma'$), antidiagonal ($m = n, \sigma = -\sigma'$), off-diagonal ($m \neq n$), and remainder cross-terms.

Off-diagonal $\sigma \neq \sigma'$. $|d\Phi/dt| \geq 2\vartheta'(t) - O(1) \rightarrow \infty$; single IBP gives $O((\log T)^{-1}) \sum_{m,n} (mn)^{-1/2} = o(U)$.

Off-diagonal $\sigma = \sigma', m \neq n$. $|d\Phi/dt| \geq \frac{1}{2} |\log(m/n)|$; $\sum_{m \neq n \leq N(T)} (mn)^{-1/2} / |\log(m/n)| = O(T^{1/2} \log T) = o(U)$.

Remainder. $|R| = O(t^{-1/4})$ cross-terms Cauchy–Schwarz to $O(T^{1/2}) = o(U)$.

Diagonal. (n, n, σ, σ) contributes, after phase expansion $\delta_\eta(t)[\vartheta'(t) - \log n] \rightarrow \eta \omega_n(t)$ with $\omega_n(t) \in [0, 1]$ for $t \geq 2\pi n^2$ by Lemma 11:

$$\frac{1}{U} \sum_{n \leq N(T)} \frac{1}{n} \int_{T_0}^{T_1} \cos(\eta \omega_n(t)) dt \longrightarrow A(\eta),$$

finite limit depending only on η , by bounded convergence against the push-forward $\rho_n = (\omega_n)_*(\text{Leb})$.

Antidiagonal. $O(N(T)/\log T) = o(U)$.

Translation invariance. Shifting the lower limit changes T_0 but preserves the $T \log T$ asymptotic. $\square$

Corollary 13 (Bochner spectral measure). *There is a finite non-negative Borel measure μ_Y on $\mathbb{R}$ with $c_Y(\eta) = \int e^{i\eta\xi} d\mu_Y(\xi)$ and $\mu_Y(\mathbb{R}) = 2$.*

Proof. c_Y is positive-definite (Cesàro limit of positive-definite kernels) and continuous at 0 ($|c_Y(\eta) - c_Y(0)|^2 \leq c_Y(0)\Delta(\eta) \rightarrow 0$ by the explicit diagonal series). Bochner’s theorem applies. $\square$

4 Band-Limit via Single Integration by Parts

Definition 14 (Windowed Fourier, PSD). $K_T(\mu) := (2\pi)^{-1} \int_0^T Z(t) \sqrt{\Theta'(t)} e^{-i\mu\Theta(t)} dt$; $S(\mu) := \limsup_T 2\pi |K_T(\mu)|^2 / \Theta(T)$.

Theorem 15 (Band-limit). *For every $c > c_*$ and every $|\mu| > 1$: $S(\mu) = 0$.*

Proof. Insert Lemma 10 into K_T and substitute $u = \Theta(t)$, so $dt = du/\Theta'(\Theta^{-1}(u))$. Mode (n, σ) of the main sum contributes

$$\mathcal{J}_{n,\sigma}(T, \mu) = \int_{u_n}^{\Theta(T)} G_{n,\sigma}(u) e^{i\Psi_{n,\sigma,\mu}(u)} du, \quad G_{n,\sigma}(u) = \Theta'(\Theta^{-1}(u))^{-1/2},$$

where $u_n := \Theta(2\pi n^2)$, and the phase is

$$\Psi_{n,\sigma,\mu}(u) = \sigma \vartheta(\Theta^{-1}(u)) - \sigma \Theta^{-1}(u) \log n - \mu u = (\sigma - \mu) \vartheta(\Theta^{-1}(u)) - \sigma \Theta^{-1}(u) \log n - \mu c \Theta^{-1}(u).$$

Phase derivative and instantaneous u -frequency. Differentiating in u via $d\Theta^{-1}(u)/du = 1/\Theta'(\Theta^{-1}(u))$,

$$\Psi'_{n,\sigma,\mu}(u) = \frac{(\sigma - \mu) \vartheta'(\Theta^{-1}(u)) - \sigma \log n - \mu c}{\Theta'(\Theta^{-1}(u))} = \sigma \omega_n(\Theta^{-1}(u)) - \mu.$$

By Lemma 11, $|\sigma \omega_n| \leq |\omega_n| < 1$, so for $|\mu| > 1$:

$$|\Psi'_{n,\sigma,\mu}(u)| \geq |\mu| - 1 =: \lambda > 0.$$

Exact cancellation in the IBP ratio. Set $x := \Theta'(\Theta^{-1}(u)) = \vartheta'(\Theta^{-1}(u)) + c$. Then $G_{n,\sigma}(u) = x^{-1/2}$ and

$$\frac{d\Psi}{du} = \frac{(\sigma - \mu)x - \sigma(c + \log n)}{x}, \quad Q_{n,\sigma,\mu}(u) := \frac{G_{n,\sigma}(u)}{d\Psi/du} = \frac{x^{1/2}}{(\sigma - \mu)x - \sigma(c + \log n)}.$$

$Q_{n,\sigma,\mu}$ is a rational function of x alone; no $\vartheta''(t)$ enters, and no bound on ϑ'' is required. As $x \rightarrow \infty$, $Q_{n,\sigma,\mu}(u) = O(x^{-1/2})$ and $(dQ_{n,\sigma,\mu}/dx) = O(x^{-3/2})$.

IBP bound. With $\beta_n := \Theta'(2\pi n^2) = \log n + c + O(n^{-4})$:

$$\int_{u_n}^{\infty} |G'_{n,\sigma}| du = \beta_n^{-1/2}, \quad \int_{u_n}^{\infty} |\Psi''_{n,\sigma,\mu}| du = \frac{c + \log n}{\beta_n},$$

and a single IBP gives

$$|\mathcal{J}_{n,\sigma}(T, \mu)| \leq \frac{C(c, |\mu|)}{\lambda} \beta_n^{-1/2} \leq \frac{C}{\lambda(\log n)^{1/2}}.$$

Remainder contribution. The Riemann–Siegel remainder contributes $\int_{T_0}^T R(t) \sqrt{\Theta'(t)} e^{-i\mu\Theta(t)} dt$; the phase slope $\mu\Theta'(t) + \arg R'(t)$ is bounded below in absolute value by $|\mu|c/2$ for large t (Lemma 10 on $\arg R'$), and IBP yields $O(T^{-1/4})$ uniformly.

Summing over modes.

$$|K_T(\mu)| \leq C \sum_{n=1}^{N(T)} \frac{n^{-1/2}}{\lambda(\log n)^{1/2}} + C = O\left(\frac{T^{1/4}}{\sqrt{\log T}}\right).$$

Since $\Theta(T) \sim \frac{T}{2} \log T$,

$$\frac{2\pi |K_T(\mu)|^2}{\Theta(T)} = O\left(\frac{1}{T^{1/2}(\log T)^2}\right) \rightarrow 0. \quad \square$$

Corollary 16 (Support of μ_Y). $\text{supp}(\mu_Y) \subseteq [-1, 1]$.

Proof. Wiener (Acta Math. 55 (1930), Thm. 29): $\mu_Y(E) = \lim_T (2\pi/\Theta(T)) \int_E |K_T(\mu)|^2 d\mu$ for bounded Borel E . Theorem 15 gives pointwise vanishing on $E \subseteq \{|\mu| > 1\}$; dominated convergence gives $\mu_Y(E) = 0$; countable additivity gives $\mu_Y(\{|\mu| > 1\}) = 0$. $\square$

5 Entire Extension and Reality of Zeros of Y

Theorem 17 (Paley–Wiener extension). *Y extends to an entire function of exponential type ≤ 1 with $|Y(z)| \leq \sqrt{2}e^{|\Im z|}$.*

Proof. Theorem (Paley–Wiener for time-average WSS functions; Yaglom, Vol. I, Thm. 4.7.3; Bochner–Phillips, Ann. Math. 43 (1942), 409–418). Let $f : \mathbb{R} \rightarrow \mathbb{C}$ be locally square-integrable and admit a time-average covariance

$$c_f(\eta) := \lim_{U \rightarrow \infty} \frac{1}{U} \int_0^U f(u) \overline{f(u + \eta)} du$$

existing for every $\eta \in \mathbb{R}$, positive-definite, and continuous at 0. Let μ_f be the Bochner spectral measure, i.e. $c_f(\eta) = \int_{\mathbb{R}} e^{i\eta\xi} d\mu_f(\xi)$, and assume $\text{supp } \mu_f \subseteq [-\tau, \tau]$. Then f extends to an entire function on $\mathbb{C}$ of exponential type $\leq \tau$ with

$$|f(z)| \leq \mu_f(\mathbb{R})^{1/2} e^{\tau|\Im z|} \quad (z \in \mathbb{C}).$$

Application. Y satisfies the hypotheses with $\tau = 1$, $\mu_f = \mu_Y$, $\mu_Y(\mathbb{R}) = 2$ (Theorem 12, Corollary 13, Corollary 16), giving $|Y(z)| \leq \sqrt{2}e^{|\Im z|}$ and exponential type ≤ 1 . $\square$

Theorem 18 (Akhiezer–Laguerre–Pólya). *If F is a real-valued entire function of exponential type $\leq \tau$ whose time-average autocorrelation measure is finite, non-negative, and supported in $[-2\tau, 2\tau]$, then F belongs to the Laguerre–Pólya class $\mathcal{LP}$ and every complex zero of F is real.*

Proof. Akhiezer, *Theory of Approximation*, §V.4; Levin, *Lectures on Entire Functions*, Lecture 16, Thm. 3. $\square$

Theorem 19 (Reality of zeros of Y). *Every complex zero of the entire extension of Y is real.*

Proof. Y is real on $\mathbb{R}$, entire of exponential type ≤ 1 (Theorem 17), with autocorrelation $\mu_Y \geq 0$, finite, supported in $[-1, 1] \subseteq [-2, 2]$ (Corollary 16). Apply Theorem 18 with $\tau = 1$. $\square$

6 The Riemann Hypothesis

Theorem 20 (Riemann hypothesis). *Every nontrivial zero ρ of ζ satisfies $\Re \rho = \frac{1}{2}$.*

Proof. Let $\rho = \beta + i\gamma$ be a nontrivial zero with $\beta \in (0, 1)$, $\gamma \in \mathbb{R}$.

Reduction to $\beta \leq \frac{1}{2}$. The functional equation $\xi(w) = \xi(1-w)$ with reality of ξ on $\mathbb{R} \cap \{\Re w = \frac{1}{2}\}$ gives the symmetry $\rho \mapsto 1 - \bar{\rho}$, so it suffices to prove $\beta = \frac{1}{2}$ assuming $\beta \in (0, \frac{1}{2}]$.

Holomorphy of Z on the strip Ω . $Z(t) = e^{i\vartheta(t)} \zeta(\frac{1}{2} + it)$ is holomorphic on $\Omega := \{t \in \mathbb{C} : |\Im t| < \frac{1}{2}\}$: ϑ is holomorphic on Ω (the poles of $\psi(\frac{1}{4} + \frac{it}{2})$ lie at $\Im t = \pm(2k + \frac{1}{2})$, $k \geq 0$), and $\frac{1}{2} + it \in \mathbb{C} \setminus \{1\}$ for every $t \in \Omega$ since the equation $\frac{1}{2} + it = 1$ has the single solution $t = -i/2 \in \partial\Omega \setminus \Omega$.

Admissible shift. The set $\{\gamma + is : s \in [0, \frac{1}{2} - \beta]\} \subset \Omega$ is compact, so ϑ' is continuous on it, and

$$M := - \inf_{s \in [0, \frac{1}{2} - \beta]} \Re \vartheta'(\gamma + is)$$

is finite. Fix any $c > \max(c_*, M)$; then $c - M > 0$ and $\Re \Theta'(\gamma + is) = \Re \vartheta'(\gamma + is) + c \geq c - M > 0$ for every $s \in [0, \frac{1}{2} - \beta]$.

Two holomorphic functions of s . Let $N := \{s \in \mathbb{C} : |\Re s| < \frac{1}{2}\}$, a connected open vertical strip of $\mathbb{C}$; for every $s \in N$, $\gamma + is \in \Omega$, so $Z(\gamma + is)$, $\vartheta(\gamma + is)$, $\Theta(\gamma + is) := \vartheta(\gamma + is) + c(\gamma + is)$, and

$\Theta'(\gamma + is) = \vartheta'(\gamma + is) + c$ are holomorphic in s . Y is entire (Theorem 17), so $Y(\Theta(\gamma + is))$ is holomorphic in $s \in N$. Define, for $s \in N$,

$$F(s) := Z(\gamma + is), \quad G(s) := \sqrt{\Theta'(\gamma + is)} Y(\Theta(\gamma + is)),$$

with $\sqrt{\cdot}$ the principal branch (domain $\mathbb{C} \setminus (-\infty, 0]$). Let $N' \subset N$ be the connected component of $\{s \in N : \Theta'(\gamma + is) \notin (-\infty, 0]\}$ containing the real segment $[0, \frac{1}{2} - \beta]$. N' is open: the real segment is a connected subset on which $\Re \Theta'(\gamma + is) \geq c - M > 0$, so $\Theta'(\gamma + is)$ lies in the open right half-plane $\{\Re w > 0\} \subset \mathbb{C} \setminus (-\infty, 0]$, a condition preserved in an open neighborhood. F and G are both holomorphic on N' . Since $0 \in N'$ and N' is open, N' contains $\{ir : |r| < \varepsilon\}$ for some $\varepsilon > 0$.

Coincidence on an imaginary segment. For $|r| < \varepsilon$, with $s = ir$, $\gamma + is = \gamma + i \cdot ir = \gamma - r \in \mathbb{R}$. Lemma 7 at the real argument $\gamma - r$ gives

$$F(ir) = Z(\gamma - r) = \sqrt{\Theta'(\gamma - r)} Y(\Theta(\gamma - r)) = G(ir).$$

1D identity theorem. $F - G$ is holomorphic on the connected open set $N' \subset \mathbb{C}$, and its zero set contains the open interval $\{ir : |r| < \varepsilon\}$, which has accumulation points in N' . Therefore $F \equiv G$ on N' ; in particular at $s = \frac{1}{2} - \beta \in [0, \frac{1}{2} - \beta] \subset N'$,

$$Z(\gamma + i(\frac{1}{2} - \beta)) = F(\frac{1}{2} - \beta) = G(\frac{1}{2} - \beta) = \sqrt{\Theta'(\gamma + i(\frac{1}{2} - \beta))} Y(\Theta(\gamma + i(\frac{1}{2} - \beta))).$$

Left side is zero. $\frac{1}{2} + i(\gamma + i(\frac{1}{2} - \beta)) = \frac{1}{2} + i\gamma - (\frac{1}{2} - \beta) = \beta + i\gamma = \rho$, and $\gamma + i(\frac{1}{2} - \beta) \in \Omega$ since $\Im(\gamma + i(\frac{1}{2} - \beta)) = \frac{1}{2} - \beta \in [0, \frac{1}{2})$, so

$$Z(\gamma + i(\frac{1}{2} - \beta)) = e^{i\vartheta(\gamma + i(\frac{1}{2} - \beta))} \zeta(\rho) = 0.$$

Consequence for Y . $\sqrt{\Theta'(\gamma + i(\frac{1}{2} - \beta))} \neq 0$ since $\Theta'(\gamma + i(\frac{1}{2} - \beta)) \in \mathbb{C} \setminus (-\infty, 0]$ (else $\frac{1}{2} - \beta \notin N'$); hence $Y(\Theta(\gamma + i(\frac{1}{2} - \beta))) = 0$. By Theorem 19, $\Theta(\gamma + i(\frac{1}{2} - \beta)) \in \mathbb{R}$.

Integral representation and conclusion. On the real segment $s \in [0, \frac{1}{2} - \beta]$, $\Theta(\gamma + is)$ is holomorphic in s , and

$$\Theta(\gamma + i(\frac{1}{2} - \beta)) - \Theta(\gamma) = \int_0^{\frac{1}{2} - \beta} \frac{d}{ds} \Theta(\gamma + is) ds = \int_0^{\frac{1}{2} - \beta} i \Theta'(\gamma + is) ds,$$

so, taking imaginary parts and using $\Theta(\gamma) \in \mathbb{R}$,

$$\Im \Theta(\gamma + i(\frac{1}{2} - \beta)) = \int_0^{\frac{1}{2} - \beta} \Re \Theta'(\gamma + is) ds \geq (\frac{1}{2} - \beta)(c - M).$$

$\Theta(\gamma + i(\frac{1}{2} - \beta)) \in \mathbb{R}$ gives the left side is 0; $c - M > 0$; the right side vanishes iff $\frac{1}{2} - \beta = 0$. Hence $\beta = \frac{1}{2}$. $\square$

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